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MyDesign Portfolio

Element C Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design requirements, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design requirements, be sure to delete the instructions. Make sure to list your design requirements in ranked order, with design requirement #1 having the highest priority. If possible, list 5 requirements that are labeled as HIGH priority and 1-5 additional requirements that are labeled as LOW priority.

Element C: Presentation and Justification of Solution Design Requirements

Signal (HIGH Priority)

- a. A working signal is important in our design because it can help let people know when the drink machine is running low so they can inform someone in case they haven't already realized it.
- b. This ties to our research because we are trying to make a sensor that can help signal people when the drink machine is low. It can work like a car signal when your gas is running low.
- c. As Mr. Jones did think it was the students causing this problem, he still thought that this was good but said we shouldn't make a signal that makes too much noise that could scare people in the lunch room. (20%)

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Can contact the people in the lunch room (HIGH Priority)

- Contacting the people in the lunch room would be a good advancement because it could let them know to fill the drink machine with more syrup and allow students to not have to worry about getting water rather than the drink they want.
- Contacting people is something that ties to our research because it can tell people who have control to do something. It likes how when your alarm goes off it goes straight to the police so they can come and help you.
- Mr. Gloster thought that this was another good idea because he knows the students aren't allowed and aren't able to put more syrup in the drink machine. (15%)

Very High Tech Device (HIGH Priority)

- We want this to be high tech and developed because we don't want it malfunctioning and for it to stop working because it will make the sensor useless.
- This requirement relates to our research because many of the design failures happen because the device stops working and it doesn't really assist people.
- Mr. Gloster said that we would probably need a lot of money to make this a high tech device. (20%)

Cost (HIGH Priority)

- To actually make this we need plenty of money because this involves a lot of advance technology which means we'd need to spend around \$150-\$200
- This doesn't really have anything to do with others but ourselves but for us the price is extremely important when making this device to benefit our school
- Mr. Jones said that we'd need to find a way to get the right materials for the device to work at all (30%)

Accuracy (HIGH Priority)

- The sensor needs to accurately know when the drink machine is running out of syrup. If it says it too late or too early it could frustrate people who are trying to refill it because they may feel like it's a waste of their time.
- Sometimes when a sensor doesn't work the way it's supposed to it can cause people to be confused and make serious mistakes.
- Mr. Jones said that accuracy would be very important when the sensor is going because it being inaccurate can cause many problems at times. (10%)

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Keeping it Clean (LOW Priority)

- Keeping it clean is just to make sure it doesn't get dirty and to make sure it doesn't cause any problem with the machine
- Sometimes too much dust can get in the sensor and cause a malfunction in the sensor and the machine.
- Mr. Gloster keeping it clean is important but it isn't needed because he thinks that the drink machine is fairly clean. (5%)

Design Requirement #7 (LOW Priority)

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.

Explain how this design requirement directly ties to your prior research.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.